

Stefan Strainovic

DATA SCIENTIST

CONTACT

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Paris, France

SKILLS

Python • **RStudio** • **SAS** • **SQL**

AWS (S3, Sagemaker, EC2)

GCP ()

Docker • Kubernetes (K8s)

GitHub • **Excel** • VBA • Power BI

dbt • PySpark

PYTHON LIBRARIES

NumPy • pandas • SciPy • Polars

matplotlib • seaborn • plotly

scikit-learn • XGBoost • LightGBM

PyTorch • TensorFlow • Transformers

SHAP • statsmodels • MLFlow • Optuna

Hyperopt • Streamlit • FastAPI • NiBabel

SpaCy • Whisper • tokenizers

LANGUAGES

English (C2 - native)

Serbian (C2 - native)

French (C1/C2 - fluent)

VOLUNTEERING

ESN Paris (2024 - Present)

- **Administrator**

ESN Caen (2023-2024)

- **Regional Representative**

EXPERIENCE

AI / ML Engineer — Stealth Startup

MAY 2025 - JANUARY 2026 — PART TIME

- **Designed a multimodal voice-based health screening model** that converts short speech recordings and their transcripts into a **clinical score** using an attention encoder architecture.
- **Combined acoustic features** (prosody, pauses, energy, spectrum) with **linguistic indicators** and **Transformer-based text embeddings** to capture both how people speak and what they say.
- **Built an end-to-end Python pipeline from raw audio to final prediction**, including preprocessing, automatic transcription, feature extraction, train / validation / test splits, cross validation and structured experiment tracking.

Machine Learning Engineer — Karmen

JULY 2024 - JANUARY 2025 — INTERNSHIP

- **Researched, designed and trained credit scoring models** on banking and financial data to support loan prioritisation, achieving high **90-percentile accuracy** on internal validation sets.
- **Implemented a full data pipeline in Python and SQL** (extraction, cleaning, aggregation, feature engineering) across thousands of transactions per client and **deployed the model in a Streamlit** app used daily by the credit team.
- **Built a client specific transaction labelling system using NLP** (TF-IDF and clustering) that reduced manual labels from several thousand to about **200 per client, speeding up file review**.
- **Prototyped an automatic bank detection system** from PDF statements, combining **CNN-based document vision** with **text** and layout cues to route each statement to the correct extraction schema and reduce parsing errors.

Statistician (Oncology) — Cyceron - CNRS

MARCH 2023 - JULY 2023 — INTERNSHIP

- **Designed and fitted multiple regression models** in R to quantify the impact of brain irradiation protocols on different T lymphocyte subtypes.
- **Structured a reproducible analysis pipeline** for raw experimental data, covering integration, cleaning, handling missing values and checking statistical assumptions.
- **Summarised results** in visualisations and interpretation notes for clinical teams, clearly highlighting main effects and study limitations.

EDUCATION

Master's in Applied Mathematics — 2024

University of Caen

Focus on **statistical modelling**, operational research, and **data management and analysis.**

Bachelor's in Mathematics — 2022

University of Paris Diderot

Focus on **fundamental and applied mathematics**.

PROJECTS & HACKATHONS

MRI 3D – Tumour Segmentation ↗ — May 2026

Implemented and compared 3D brain tumour segmentation architectures (**3D U-Net**, **Attention U-Net**, **Swin UNETR**) on MRI volumes, with patch-based training, sliding-window inference, and quantitative evaluation using Dice and Hausdorff distance metrics.

Genome – Bacteriophage (hackathon) — May 2025

Developed a genomics pipeline to classify bacteriophage genome segments into structural regions by encoding sequences and applying a Transformer style model to predict the structural class of each segment.

MRI – Tumour Classification — February 2024

Implemented a CNN with transfer learning (VGG style) for brain tumour classification on MRI, including preprocessing, data augmentation, training and test evaluation, reaching around 98% accuracy.

ESN Paris (2024 - Aujourd'hui)

- **Administrateur**

ESN Caen (2023-2024)

- **Représentant régional**

BÉNÉVOLAT

COMPÉTENCES

Python · **R**-studio · **SAS** · **SQL**
AWS (S3, Sagemaker, EC2)
Docker · **Kubernetes**
Github
Excel · VBA
Power BI

LIBRARIES PYTHON

NumPy · pandas · SciPy · Polars
matplotlib · seaborn · plotly
scikit-learn · XGBoost · LGBM
PyTorch · TensorFlow · Transformers
SHAP · statsmodels · MLFlow · Optuna
Hyperopt · Streamlit · FastAPI · NiBabel
SpaCy · Whisper · tokenizers

LANGUES

Anglais (C2 - langue maternelle)
Serbe (C2 - langue maternelle)
Français (C1/C2 - courant)

BÉNÉVOLAT

Built a **3D tumour segmentation model based on nnU-Net** with an attention block to delineate tumours on MRI volumes, with volume normalisation, patch extraction, training and qualitative and quantitative evaluation of predicted masks.

MRI 3D – Tumour Segmentation (

Implemented and compared 3D brain tumour segmen
Attention U-Net, Swin UNETR) on MRI volumes, with p
inference, and quantitative evaluation using Dice and H

Genome – Bacteriophage (hackatl

Developed a genomics pipeline to classify bacteri
structural regions by encoding sequences and applyin
predict the structural class of each segment.

- **Créé un modèle d'IA** qui transforme de courts enreg
transcriptions en score de santé, en fusionnant feature
audio et texte dans un encoder attention. (add the dor
- **Mis en place un pipeline** complet audio → prédiction
features, transcription, split train/val, cross-validation,